

Problem Set 6

Submission:

Friday, 10/04/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.

Only problems marked with (*) will be graded.

1. Expectation, variance and higher moments

- (a) Consider a continuous real random variable X with law characterized by the probability density function

$$f_X(x) = \begin{cases} \frac{c}{(1+x)^5}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

- Determine the value of c .
 - Calculate the cumulative distribution function of X .
 - Determine the values $\mathbf{E}[X]$ and $\text{Var}[X]$.
Hint: First calculate $\mathbf{E}[1 + X]$ and $\mathbf{E}[(1 + X)^2]$.
 - What is the largest $n \in \mathbb{N}$ such that $\mathbf{E}[X^n]$ exists? Justify your answer.
- (b) A continuous random variable X has a Γ -distribution with parameters $\alpha > 0$ and $\lambda > 0$, written as $X \sim \Gamma(\alpha, \lambda)$, if its law is characterized by the probability density function

$$f_X(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-\lambda x}, & x > 0, \\ 0, & x \leq 0, \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$.

- Show that f_X defined above is a density function.
- Show by integration by parts that for all $\alpha > 0$, $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$. In particular, since $\Gamma(1) = 1$, conclude that $\Gamma(n) = (n-1)!$ for all $n \in \mathbb{N}$.
- Calculate $\mathbf{E}[X]$ if $X \sim \Gamma(\alpha, \lambda)$, where $\alpha, \lambda > 0$.
- Calculate $\text{Var}[X]$ if $X \sim \Gamma(\alpha, \lambda)$, where $\alpha, \lambda > 0$.

2. Properties of the expectation

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space.

- (a) Let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{N}, \mathcal{P}(\mathbb{N}))$ be a (discrete) $\mathbb{N}$ -valued random variable. Show that

$$\mathbf{E}[X] = \sum_{n=1}^{\infty} \mathbf{P}[X \geq n],$$

where we assume that $\mathbf{E}[X] < \infty$. Use this to calculate $\mathbf{E}[X]$ if $X \sim \text{Geo}(p)$, $p \in (0, 1)$.

Hint: Explain first why for nonnegative numbers $(a_{n,k})_{n,k \in \mathbb{N}}$, one has that $\sum_{k=1}^{\infty} \sum_{n=1}^{\infty} a_{n,k} = \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} a_{n,k}$.

- (b) If $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a non-negative real random variable (i.e. $X(\omega) \geq 0$ for every $\omega \in \Omega$), then one can show similarly to the above formula that

$$\mathbf{E}[X] = \int_0^\infty (1 - F_X(x)) dx,$$

where we assume that $\mathbf{E}[X] < \infty$. Use this formula to calculate $\mathbf{E}[X]$ for $X \sim \mathcal{E}(\lambda)$ for $\lambda > 0$.

3. (*) **Hazard rate**

[4 Points]

We model the life-time of a device by a continuously distributed random variable $T \geq 0$ with cumulative distribution function $F(t)$ and probability density function $f(t)$. We assume that f is continuous in every $t \geq 0$. The *hazard rate* is defined as

$$\lambda(t) = \lim_{h \searrow 0} \frac{\mathbf{P}[t \leq T \leq t+h | T \geq t]}{h}, \quad t \geq 0$$

(this is also relevant for life-insurance purposes).

- (a) Express $\lambda(t)$ in terms of $f(t)$ and $F(t)$.
- (b) Suppose that $T \sim \mathcal{E}(\alpha)$ with $\alpha > 0$, and calculate $\lambda(t)$.
- (c) Verify that $F(t) = 1 - \exp\left(-\int_0^t \lambda(s) ds\right)$, when $t \geq 0$.
- (d) Calculate $F(t)$ and $f(t)$ for $t \geq 0$ when $\lambda(t) = \alpha t^\gamma$ ($\alpha > 0, \gamma > 0$) (Weibull distribution).
- (e) For $T \geq 0$, show that $\mathbf{E}[T] = \int_0^\infty \exp\left(-\int_0^t \lambda(s) ds\right) dt$.

4. (*) **More on expectation**

[4 points]

- (a) Let $\mathcal{S}_n$ be set of all permutations of $\{1, \dots, n\}$ and $\mathbf{P} = \mathcal{U}(\mathcal{S}_n)$ be the uniform distribution on $\mathcal{S}_n$. For every permutation $\omega \in \mathcal{S}_n$, let $X(\omega)$ be the number of fixed points of ω . Find $\mathbf{E}[X]$.
Hint: Consider for $i = 1, \dots, n$, $A_i = \{\omega \in \mathcal{S}_n : \omega(i) = i\}$. Then $X = \sum_{1 \leq i \leq n} \mathbf{1}_{A_i}$.
- (b) In a fictional company, a worker's legal code specifies as a holiday any day during which at least one worker has their birthday. All other days are working days. How many workers must the factory employ so that the expected number of working worker-days is maximized during the year? (Worker-days = Headcount $\times$ Days.)
Hint: First consider the expectation of the total number of birthday-free days.